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10 CFR 50.73

GNRO-2020/00005

March 4, 2020

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

SUBJECT: Grand Gulf Nuclear Station, Unit 1, Revised Licensee Event Report
2019-003-01, Secondary Containment Personnel Door Failure

NRC Docket No. 50-416
Renewed Facility Operating License No. NPF-29

Attached is a revised Licensee Event Report 2019-003-01, Secondary Containment Personnel Door Failure. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(v)(C) for an event or condition that at the time of discovery could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material.

This letter contains no new commitments. If you have any questions or require additional information, please contact Jim Shaw at 601-437-2103.

Sincerely,

A handwritten signature in black ink, appearing to read "E. A. Larson".

Eric A. Larson
EAL/fas

Attachment: Licensee Event Report 2019-003-01

cc: NRC Region IV - Regional Administrator
NRC Senior Resident Inspector, Grand Gulf Nuclear Station
NRR Project Manager

GNRO-2020/00005

Attachment
Licensee Event Report 2019-003-01



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form <http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name Grand Gulf Nuclear Station, Unit 1	2. Docket Number 05000 416	3. Page 1 OF 3
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4. Title
Secondary Containment Personnel Door Failure Due to Broken Hinges

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
08	05	2019	2019	- 003	- 01	03	04	2020	N/A	05000 N/A
									Facility Name	Docket Number
									N/A	05000 N/A

9. Operating Mode 1	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
10. Power Level 100	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(iii)
☐ 50.73(a)(2)(i)(C) ☐ Other (Specify in Abstract below or in NRC Form 366A)				

12. Licensee Contact for this LER

Licensee Contact Jim Shaw, Manager Regulatory Assurance	Telephone Number (Include Area Code) (601) 437-2103
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To ICES	Cause	System	Component	Manufacturer	Reportable To ICES
B	NH	NH	Overly	Y					

14. Supplemental Report Expected

☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No

15. Expected Submission Date

Month	Day	Year
NA	NA	NA

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On August 5, 2019, at 09:36, the Control Room was notified that a Secondary Containment Door was found with door hinges broken and the door was unable to be closed.

Grand Gulf Nuclear Station performed action A.1 for Technical Specification (TS) LCO 3.6.4.1, Secondary Containment, not met due to the Secondary Containment door being inoperable. The broken hinges were replaced with qualified spares, and TS LCO 3.6.4.1 was met on August 5, 2019 at 13:36.

Based on vendor's engineering analysis, the Direct Cause was Door 1T101A401B hinge pins failed due to brittle fracture, resulting in a loss of the Secondary Containment Safety Function.

The Root Cause for the event is a design deficiency related to the mounting of the door hinges. The planned corrective action is a modification to the door. This will be done when secondary containment is not required to be operable.

This report is made pursuant to 10 CFR 50.73(a)(2)(v)(C) for an event or condition that at the time of discovery could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material. There were no consequences to the general safety of the public, nuclear safety, industrial safety and radiological safety for this event.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Grand Gulf Nuclear Station, Unit 1	05000-416	YEAR 2019	SEQUENTIAL NUMBER - 003	REV NO. - 01
NARRATIVE				
Plant Conditions:				
Grand Gulf Nuclear Station (GGNS) Unit 1 was operating at approximately 100 percent power in MODE 1. There were no structures, systems, or components that were inoperable that contributed to the event.				
Description:				
On August 5, 2019, at 09:36, the Control Room was notified that the Secondary Containment Door (NF) 1T101A401B (or 1A401B) was found with door hinges broken and the door was unable to be closed.				
With the Secondary Containment Door unsecured, Secondary Containment was declared INOPERABLE and Secondary Containment Technical Specification LCO 3.6.4.1 was not met. TS LCO 3.6.4.1 was met on August 5, 2019 at 13:36 and Secondary Containment was declared operable.				
Door 1A401B is a security door, and the on-shift Security Force initiated compensatory measures immediately. Therefore, the Security Plan was not affected.				
Door 1A401B is a fire barrier. Technical Requirements Manual (TRM) 6.2.8 Condition A was not met. Operations declared the door NON-FUNCTIONAL and established an hourly fire watch per TRM 6.2.8 Action A.2.2.				
TS LCO 3.6.4.1, and TRM 6.2.8 function were met when maintenance technicians replaced the broken hinges for the door and the door was declared operable on August 5, 2019, at 13:36.				
Reportability:				
This event was reported as an event or condition that, at the time of discovery could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material per 10 CFR 50.72(b)(3)(v)(C). The event notification is EN 54201.				
This event is also reportable per 10 CFR 50.73(a)(2)(v)(C).				
Direct Cause:				
Based on vendor's engineering analysis, the Direct Cause was Door 1T101A401B hinge pins failed due to brittle fracture, resulting in a loss of the Secondary Containment Safety Function.				
Root Cause:				
The Root Cause for this event is the unique design of the hinges, which introduces a side-loaded stress on the hinge pins while the door is open. The hinges on the Door 1A401B are surface-mounted cam-lift hinges that are designed to be installed in a full-mortised configuration.				
Immediate Corrective Actions:				
Security Force members initiated compensatory measures immediately while Door 1A401B was unable to perform its security barrier function, satisfying the Security Plan requirements.				



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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Grand Gulf Nuclear Station, Unit 1		05000-416	YEAR	SEQUENTIAL NUMBER	REV NO.
			2019	- 003	- 01

Operators and Security Force members were also serving as an hourly fire watch while Door 1A401B was unable to perform its fire barrier function, satisfying TRM 6.2.8 Action A2.2.2.

Work was performed to replace all of the hinges restoring Door 1A401B to its design configuration.

Planned Corrective Actions to Preclude Recurrence, which may be modified per the Entergy corrective action program:

The Corrective Actions for this event will be implementing a modification to install a more robust hinge design in Door 1A401B.

Safety Significance:

There were no actual consequences for this event. There was no radiological release from the Secondary Containment as a result of this event. There were no other actual consequences to safety of the general public, nuclear safety, industrial safety and radiological safety.

Previous Similar Event:

A three-year review of similar events was conducted.

On July 9, 2017, a pressure boundary door to the Control Room Envelope was left unsecured for approximately one minute. The Control Room Envelope was declared inoperable, due to the open boundary door. The direct cause of the event was that the door was degraded, resulting in the door dragging on the floor such that the automatic door closure device would not function properly. The event was reported as a Loss of Safety Function per 10 CFR 50.73(a)(2)(v)(D).
LER 05-416/2017-005-00.

The Corrective Actions from this event implemented a maintenance strategy, which did not include the Secondary Containment Door that is the subject of this LER.

GGNS has planned corrective actions to address the maintenance strategy associated with the Secondary Containment Doors.